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X-JEPA: A Novel Joint Learning Cross-Modal Predictive Alignment Framework for Remote Sensing Image Retrieval

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Abstract

The growing scale and heterogeneity of remote sensing (RS) imagery demand robust, scalable frameworks for content-based image retrieval across sensor modalities. We introduce X-JEPA*, a novel predictive self-supervised architecture explicitly designed for cross-modal remote sensing image retrieval (RS-CMIR), and the first to extend joint embedding predictive paradigms beyond unimodal domains. Unlike prior contrastive or reconstruction-based methods,* X-JEPA *formulates representation learning as a latent forecasting task: predicting the semantic embedding of a target modality given context from another. To enforce modality-invariant alignment, we propose a geometry-aware Prediction Space Alignment (PSA) loss, which captures the structure of the latent space without requiring pixel-level reconstruction or modality pairing. We evaluate* X-JEPA *on two large-scale benchmarks—BEN-14K (Sentinel-1/Sentinel-2) and fMoW (RGB/Sentinel) across both unimodal and cross-modal retrieval tasks.* X-JEPA *consistently outperforms state-of-the-art self-supervised baselines, including MAE, SatMAE, CrossMAE, CSMAE-SESD, CROMA, SkySense, DeCUR, and REJEPA, achieving up to 11.0% F1-score improvement in cross-modal retrieval and 9.8% in unimodal settings. Despite its high retrieval accuracy, the model remains lightweight, requiring fewer parameters and yielding 8–10% F1-score gains on average, establishing a new state-of-the-art for scalable, sensor-agnostic RS-CMIR.* [1]

1. Introduction

The proliferation of satellite constellations has produced vast, multi-modal remote sensing (RS) archives, featuring a rich diversity of data from optical, Synthetic Aperture Radar (SAR), and multispectral sensors. This data explosion creates a critical need for effective Cross-Modal Image Retrieval (CMIR) systems technologies, capable of finding semantically related scenes across different sensor types, such as identifying a specific urban area in an optical image and retrieving its corresponding view in a SAR image. Such capabilities are indispensable for applications like longitudinal land use monitoring, disaster response, and environmental science, which depend on integrating information from disparate data sources to form a comprehensive understanding of a geographic region.

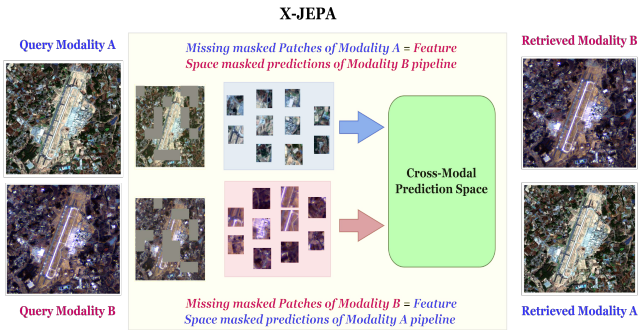


Figure 1. Overview of X-JEPA. Given a query from one modality, the model predicts masked feature representations of the opposite modality in latent space using a cross-modal context. Retrieval is performed by aligning the predicted and target embeddings within a shared semantic prediction space.

However, RS-CMIR is a fundamentally challenging task due to the significant modality gap: the inherent and profound differences in how sensors capture information. Optical sensors record reflected sunlight, producing images sensitive to spectral color, illumination, and atmospheric conditions. In contrast, active sensors like SAR measure microwave backscatter, revealing surface texture, structure, and moisture content, but are also subject to speckle noise and geometric distortions unique to their side-looking acquisition geometry. This physical divergence means that semantically identical scenes often appear visually uncorrelated, rendering direct comparison of raw pixel data ineffective and complicating the alignment of high-level semantic features.

[1]Code and pretrained models will be released upon acceptance.

Existing approaches have struggled to bridge this modality gap. Supervised methods [9, 13], while performant, are constrained by their reliance on large-scale, manually annotated datasets of co-registered image pairs. Creating such datasets is prohibitively expensive and does not scale to the global diversity of sensors, geographic locations, and temporal variations, creating a significant scalability and generalization gap. Self-supervised learning (SSL) offers a more scalable alternative. However, dominant SSL paradigms [4, 7, 39, 41] exhibit critical limitations in the cross-modal RS context. Contrastive learning [10, 36], which learns by pulling "positive" pairs closer and pushing "negative" pairs apart, often fails because visually dissimilar RS images can be semantically identical (e.g., a flooded field vs. a dry one). This ambiguity leads to a "false negative" problem, causing semantic drift in the learned representations. On the other hand, generative masked image modeling (MIM) techniques like Masked Autoencoders (MAEs) [1, 24] focus on reconstructing low-level, modality-specific details such as pixel colors or speckle patterns. This focus on surface-level fidelity inhibits the development of the abstract, modality-invariant semantic features required for effective CMIR, creating a semantic abstraction gap.

To overcome these limitations, we introduce X-JEPA (Cross-modal Joint Embedding Predictive Architecture), a novel SSL framework designed specifically for RS-CMIR (see Figure 1). Inspired by the predictive principles of JEPA [2, 26] and extending its unimodal RS application in REJEPA [14], X-JEPA is a new prwdictive paradigm to leverage JEPA-style predictive learning for cross-modal retrieval. Instead of reconstructing pixels or contrasting explicit pairs, our model learns a deeper semantic correspondence by predicting the latent representation of a target modality using contextual information from a source modality. This approach inherently bypasses modality-specific details and focuses directly on high-level, shared semantic information.

Our primary contributions to address the unique challenges of the RS modality gap are :

- We introduce a novel predictive-based architecture for the task of cross-modal remote sensing retrieval, specifically engineered to bridge the semantic gap between different sensor modalities by predicting latent representations rather than raw pixels.
- We propose the use of learnable shared queries within the predictor module, as a mechanism to create semantic anchors, enabling the model to learn a consistent and discriminative set of features that are coherent across both optical and SAR domains.
- We design a novel Prediction Space Alignment (PSA) loss, a mutual information-guided objective that enforces geometric consistency in the latent prediction space, leading to more discriminative and modality-invariant representations than what is achievable with conventional SSL losses.
- To ensure the learned representations are rich and informative, we integrate Variance-Invariance-Covariance Regularization (VICReg) [5], which prevents representational collapse by enforcing variance and decorrelation within the embeddings.

We validate the effectiveness of X-JEPA through extensive experiments on large-scale RS-CMIR benchmarks, demonstrating state-of-the-art performance.

2. Related Works

2.1. Self-Supervised Learning (SSL) in RS

SSL has become a cornerstone in remote sensing (RS) research, enabling the learning of semantically rich and transferable representations from large-scale unlabeled data [41].

Contrastive learning has laid the groundwork for early SSL in RS, with frameworks like SimCLR [10], MoCo [23], BYOL [20], Barlow Twins [45], and SimSiam [11] adapted using geometric augmentations, spatial priors, and temporal alignment. DINO [8] introduced self-distillation with momentum encoders and multi-crop views, enabling learning without explicit negatives. In RS, contrastive techniques have been refined to handle spatial complexity and multimodal variability. GLCNet [27] applies contrastive pretraining for semantic segmentation, while SeCo [30] leverages seasonal alignment for invariant feature learning.

Masked image modeling (MIM) has emerged as a promising alternative, learning features by reconstructing masked regions instead of comparing pairs. MAE [24] and SimMIM [43] introduced efficient transformer-based MIM frameworks, which have been extended to RS with domain-specific variants like SatMAE [16], ScaleMAE [32], SatMAE++ [31], RingMo [37], and CSMAE [22]. DeCUR [42] proposes a redundancy-aware MIM framework that reconstructs token-level features while enforcing latent decorrelation, making it suitable for dense RS scenes. While effective in unimodal settings, MIM methods operate in pixel space and are sensitive to modality-specific texture patterns, making them less suited for semantically aligning heterogeneous modalities with differing visual characteristics.

2.2. Cross-Modal Representation Learning in RS

Cross-modal representation learning in RS aims to unify heterogeneous sensor modalities within a shared semantic space, such as optical, SAR, and multispectral. Traditional image retrieval methods typically rely on supervised training over large annotated multi-modal datasets, often requiring land-cover labels to enforce semantic alignment. However, collecting such annotations is labor-intensive and difficult to scale. Prior work has addressed this through deep hashing [12], multi-stage training [44], and synthetic data generation [25], but these approaches remain constrained by their dependence on labeled multi-sensor data, limiting

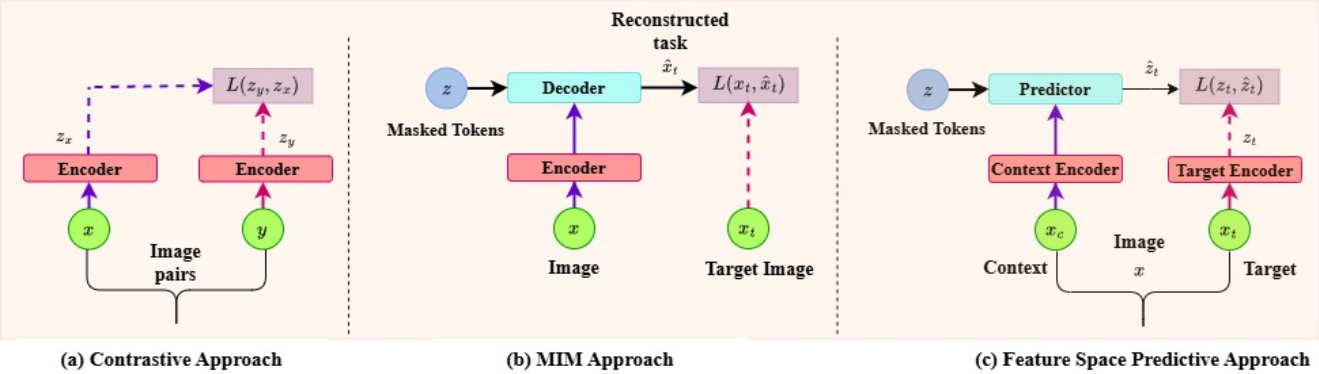


Figure 2. Overview of self-supervised learning paradigms adapted from [28] based on their core training objectives. (a) **Contrastive Learning** drives representation alignment by pulling together embeddings (z_x, z_y) of positive pairs while pushing apart negatives, often relying on augmentations. (b) **Masked Image Modeling (MIM)** predicts masked pixel regions $\hat{x}_t$ from partially observed inputs using a decoder, focusing on low-level reconstruction. (c) **Feature Prediction** (e.g., X-JEPA) learns by estimating target representations $\hat{z}_t$ from contextual features z_c, enabling abstraction directly in latent space without reconstruction or explicit contrast.

their scalability in real-world RS scenarios. To mitigate this, self-supervised approaches have gained momentum in RS-CMIR by minimizing reliance on annotated datasets. Contrastive methods like SS-CMIR [36] and CCLS2T [46] aims to align representations by maximizing mutual information between co-located image pairs. However, their dependence on spatial alignment reduces robustness under distributional shifts or sensor misalignment. CSMAE [22] extends masked autoencoders to the cross-modal setting by introducing both intra- and inter-modal reconstruction losses. While incorporating mutual information constraints such as LMIM, its reconstruction-driven nature limits the ability to capture deeper semantic structure in the latent space. More recent methods like CROMA [19], SkySense [21] explore scalable pretraining over heterogeneous modalities by learning shared semantic representations without strict modality alignment.

2.3. Feature-Space Predictive Learning

Predictive learning in latent space has emerged as a powerful SSL paradigm, shifting focus from pixel reconstruction to semantic abstraction and enabling learning from partial context inputs. Joint Embedding Predictive Architectures (JEPAs) [26], based on Siamese encoder-predictor designs, exemplify this trend by inferring abstract representations from contextual inputs without relying on pixel-level targets or handcrafted alignment strategies. Originally proposed for unimodal settings, JEPA-style models have shown strong generalization and training efficiency across domains, including vision [2], audio [18], video [6], graphs [34], and point clouds [33]. In the RS domain, SAR-JEPA [28] adapts JEPA principles to SAR imagery by integrating gradient-guided encoders, improving resilience to structural noise. REJEPA [14] is the first to explore JEPA-style predictive learning in RS retrieval, but it is confined to the unimodal setting and does not address cross-modal alignment. AnySat [3] proposes a unified architecture for multi-resolution fusion across sensors but does not formulate learning as a predictive forecasting task, nor does it address semantic alignment in latent space.

Key Highlights: Despite their success in unimodal contexts, JEPA-based models have not yet been extended to cross-modal representation learning. This presents a significant gap in predictive SSL, particularly for retrieval tasks in RS where modality-invariant features are crucial. Pixel-level reconstruction methods often overfit to modality-specific textures and require spatial alignment, which is unreliable across heterogeneous RS sensors. To address this gap, we propose X-JEPA, the novel cross-modal extension of JEPA for RS retrieval. It introduces three key contributions: (i) cross-attentional encoder-predictors for dynamic multi-modal fusion, (ii) shared learnable queries that serve as semantic anchors across modalities, and (iii) a PSA loss that enforces structure-aware alignment without relying on handcrafted pairs or pixel-level decoding.

3. Proposed Methodology

3.1. Problem Definition

Let $\mathcal{X} = \{x_i^{(a)}\}_{i=1}^N$ and $\mathcal{Y} = \{x_i^{(b)}\}_{i=1}^N$ denote two sets of images from heterogeneous RS modalities a and b (e.g., BEN (S1↔S2) or fMoW (RGB↔Sentinel)), where each $x_i^{(a)}$ and $x_i^{(b)}$ represent semantically corresponding scenes captured by different sensors. The objective in remote sensing cross-modal image retrieval (RS-CMIR) is to retrieve the most semantically relevant images from modality b given a query image from other modality$x_q^{(a)} \in \mathcal{X}$, and vice versa, without relying on human annotations or pixel-level correspondence. To address this task, we propose X-JEPA, a predictive joint embedding architecture designed to learn semantically aligned representations across modalities. The model learns to forecast the masked latent repre-

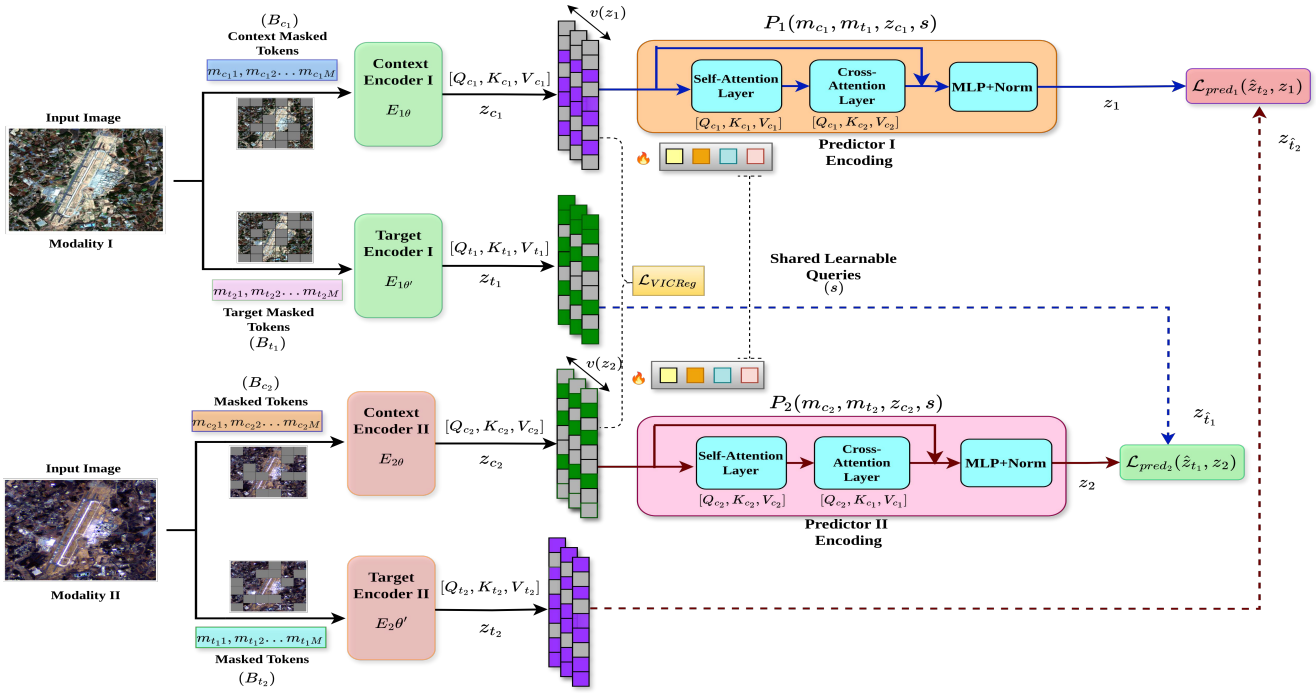


Figure 3. Overview of the proposed X-JEPA architecture for cross-modal remote sensing image retrieval (RS-CMIR). For each modality, masked input tokens are passed through a context encoder E_θ and a target encoder $E_{\theta'}$ to obtain contextual representations (z_{c_1}, z_{c_2}) and target features (z_{t_1}, z_{t_2}). Each context feature is processed by a modality-specific predictor (P_1 or P_2), consisting of a self-attention layer, a cross-attention layer, and an MLP with layer normalization. Learnable tokens between predictors enable bidirectional semantic exchange across modalities. The predicted representations $\hat{z}_{t_1}$ and $\hat{z}_{t_2}$ are aligned with their cross-modal targets z_{t_2} and z_{t_1} respectively, using the combined predictor training objectives $\mathcal{L}_{\text{pred}_1}(\hat{z}_{t_1}, z_2)$ and $\mathcal{L}_{\text{pred}_2}(\hat{z}_{t_2}, z_1)$. A regularization term $\mathcal{L}_{\text{VICReg}}$ is applied on the context branch across modalities to promote representation diversity and prevent collapse.

sentation $z_t^{(b)}$ of a target image $x_t^{(b)}$ by conditioning on the context representation $z_c^{(a)}$ extracted from a corresponding source image $x_c^{(a)}$. This prediction is performed in the latent space as :

$$\hat{z}_t^{(b)} = P_\phi(z_c^{(a)}, m),$$

where $z_c^{(a)} = E_\theta(x_c^{(a)})$ and $z_c^{(b)} = E_\theta(x_c^{(b)})$ denote the context embeddings from modalities a and b, respectively; $z_t^{(a)} = E_{\theta'}(x_t^{(a)})$ and $z_t^{(b)} = E_{\theta'}(x_t^{(b)})$ are the target embeddings; and m denotes a set of learnable mask tokens encoding spatial and modality-specific context. During inference, retrieval is performed by identifying the nearest neighbor in modality b for a given query from modality a based on the learned feature representations.

3.2. Cross-Modal Predictive Framework (X-JEPA)

The framework is structured on three principal components:

A. Modality-Aware Dual Encoders $(E_\theta, E_{\theta'})$

The architecture begins with dual encoders that process paired views with non-overlapping spatial masks. Each encoder applies a non-overlapping masking strategy to separate visible context patches and masked target regions. While the two encoders share the same backbone structure in a Siamese configuration, they operate independently to capture modality-specific statistics. Given inputs $x_c^{(a)}$ and $x_t^{(b)}$ from modalities a and b, each encoder generates latent representations from context views $(E_\theta(x_c^{(a)}), E_\theta(x_t^{(b)}))$, and target views as $(E_{\theta'}(x_c^{(a)}), E_{\theta'}(x_t^{(b)}))$, which are then used for prediction.

B. Cross-Attentional Predictor (P_ϕ)

The prediction module forms the core of X-JEPA cross-modal alignment. It receives the context features $z_c^{(a)}$, a set of routed mask tokens $m^{(a)}$, and a set of learnable queries s acting as modality-agnostic anchors that facilitate consistent information exchange across encoders. Using a cross-attentional design, the predictor estimates the masked target features $\hat{z}_t^{(b)} = P_\phi(z_c^{(a)}, m^{(a)}, s)$, learning to bridge semantic gaps between heterogeneous inputs without relying on paired supervision.

C. Shared Queries Routing

A key architectural innovation in X-JEPA is the introduction of learnable shared queries s across the predictor modules, which facilitates semantic interaction across modali-

ties. These queries are integrated into the attention mechanism alongside masked tokens and context embeddings from the opposite views. It enables the model to transfer high-level semantics across sensor domains without requiring paired supervision or pixel-level correspondence.

3.3. Disjoint Context-Target Encoding

Context Encoder (E_θ) : Given an input image x, a binary mask B_c is applied to retain only visible patches. These masked visible tokens, denoted as x_c, are passed through the context encoder E_θ. The resulting output is a set of latent tokens:

$$z_c = E_\theta(x_c) = \{z_{c1}, z_{c2}, \ldots, z_{cM}\} \tag{1}$$

Here, z_c encodes the semantic information from the visible regions of the context view and serves as the conditioning input for predicting the masked target features.

Target Encoder ($E_{\theta'}$) : The target encoder $E_{\theta'}$ receives the same input image x as the context encoder but applies a disjoint masking pattern B_t derived from the opposite modality. This cross-modal masking ensures that the model predicts semantic regions influenced by a different sensor's spatial structure. The separation between B_t and the context mask B_c guarantees that no information leakage occurs between branches, enforcing a stricter predictive constraint. The target encoder outputs a set of latent tokens:

$$\hat{z}_t = E_{\theta'}(x_t) = \{z_{t1}, z_{t2}, \ldots, z_{tN}\} \tag{2}$$

where x_t is the masked version of x, constructed with a disjoint mask derived from the opposite modality B_t. These tokens represent the semantic targets that the model learns to forecast using context features from the other modality.

3.4. Cross-Modal Semantic Prediction Module

The novel cross-modal prediction module P_ϕ introduced in X-JEPA facilitates semantic alignment across modalities by estimating the masked feature representations $\hat{z}_t^{(a)}$ in modality a, conditioned on contextual signals from both modalities.

Given a query modality a, the module receives the following inputs: (i) context embeddings $z_c^{(a)}$ from the visible regions of modality a, (ii) routed mask tokens $[m^{(a)}, m^{(b)}]$ corresponding to the masked regions of modality a and b respectively, and (iii) learnable shared queries s that act as a modality-agnostic semantic anchor.

$$z_{concat}^{(a)} = concatenate(s, z_c^{(a)}) \tag{3}$$

$$h_1 = \text{SelfAttn}(z_{concat}^{(a)}) + z_{concat}^{(a)} \tag{4}$$

$$h_2 = \text{CrossAttn}(h_1, z_c^{(b)}) + h_1 \tag{5}$$

$$z_t^{(a)} = \text{MLP}(h_2) + m^{(a)} \tag{6}$$

Here, $z_c^{(a)}$ acts as the *query* input to the cross-attention block, while the context embeddings from the opposite modality $z_c^{(b)}$ form the *keys* and *values*. The output $z_t^{(a)}$ represents the predicted feature embedding of the masked tokens in modality a and is compared with the actual target embedding $z_t^{(b)}$ from modality b using the latent alignment objective. This asymmetric prediction enforces semantic correspondence between modalities in the latent space, allowing the model to bridge heterogeneous sensor views without relying on paired supervision.

3.5. Training Objective

X-JEPA is trained using a two-part objective that combines predictive supervision in the latent space with statistical regularization to ensure robust and discriminative cross-modal representation learning.

1. Predictive Alignment Loss : The first component, $\mathcal{L}_{\text{pred}}$, enforces semantic alignment between paired modalities. It consists of:

- **L2 Loss** : A standard mean squared error term for direct prediction accuracy:

$$\mathcal{L}_{\text{L2}} = \frac{1}{M} \sum_{i=1}^{M} \|z_{ti}^{(a)} - z_{ti}^{(b)}\|_2^2 \tag{7}$$

- **Prediction Space Alignment (PSA) Loss** : A geometry-aware latent alignment loss that dynamically adapts to the underlying feature distribution structure, enabling fine-grained alignment across modalities in the learned representation space:

$$\mathcal{L}_{\text{PSA}} = \frac{1}{M} \sum_{i=1}^{M} \sqrt{(z_{ti}^{(a)} - z_{ti}^{(b)})^\top \mathbf{M} (z_{ti}^{(a)} - z_{ti}^{(b)})} \tag{8}$$

  where $\mathbf{M} \in \mathbb{R}^{d \times d}$ is a positive definite matrix that captures feature-space in a modality-agnostic manner.

The combined prediction loss :

$$\mathcal{L}_{\text{pred}} = \mathcal{L}_{\text{L2}} + \lambda_{PSA} \cdot \mathcal{L}_{\text{PSA}} \tag{9}$$

where λ_{PSA} controls the contribution of the PSA loss.

2. VICReg-Based Regularization : To prevent feature collapse and promote diversity, we adopt the Variance-Invariance-Covariance Regularization (VICReg) [5]:

- **Invariance Loss**

$$\mathcal{L}_{\text{inv}} = \frac{1}{n} \sum_{i=1}^{n} \|z_c^i - z_t^i\|_2^2 \tag{10}$$

- **Variance Loss**

$$\mathcal{L}_{\text{var}} = \frac{1}{d} \sum_{j=1}^{d} \max(0, \gamma - \sqrt{\text{Var}(Z_{:,j}) + \epsilon}) \tag{11}$$

- **Covariance Loss**

$$\mathcal{L}_{\text{cov}} = \frac{1}{d} \sum_{i \neq j} [\text{Cov}(Z)]_{i,j}^2 \tag{12}$$

Table 1. F1-score (%) comparison of self-supervised learning approaches on unimodal and cross-modal remote sensing image retrieval tasks. Evaluations are conducted on two benchmarks—BEN-14K and fMoW.

Methods	Params (M)	BEN-14K				fMoW			
		unimodal		Cross-modal		unimodal		Cross-modal	
		S1→S1	S2→S2	S1→S2	S2→S1	RGB→RGB	SEN→SEN	RGB→SEN	SEN→RGB
MAE [24]	224.87	60.81	72.04	41.78	46.12	58.73	61.77	34.23	35.12
MAE-RVSA [40]	227.75	55.40	71.47	36.66	38.05	55.26	60.28	40.46	38.65
SatMAE [16]	329.40	70.86	78.71	49.57	52.48	61.85	56.63	50.21	51.05
SatMAE++ [31]	329.14	67.29	76.48	50.21	54.98	60.09	57.75	50.39	50.03
ScaleMAE [32]	284.35	62.73	NA	NA	NA	64.26	69.56	51.52	53.45
CrossMAE [38]	250.57	66.45	71.28	49.46	48.71	66.71	63.46	48.21	51.78
CSMAE-SESD (Disjoint) [22]	210.64	70.62	39.01	38.74	38.42	51.56	41.09	30.87	30.84
SkySense [21]	398.04	69.87	73.42	50.26	52.11	68.81	67.45	51.62	52.41
CROMA [19]	310.54	68.48	72.71	46.53	48.61	65.23	66.73	49.71	47.68
DeCUR [42]	250.54	71.26	75.36	40.78	41.83	63.42	60.17	48.12	45.43
REJEPA [14]	197.09	**76.38**	75.42	55.46	56.32	73.53	75.87	50.08	49.85
X-JEPA (Ours)	**172.86**	72.98	**82.65**	**61.23**	**63.73**	**75.26**	**77.22**	**53.11**	**54.40**

The total VICReg loss is :

$$\mathcal{L}_{\text{VICReg}} = \lambda_i \cdot \mathcal{L}_{\text{inv}} + \lambda_v \cdot \mathcal{L}_{\text{var}} + \lambda_c \cdot \mathcal{L}_{\text{cov}} \quad (13)$$

where $\lambda_i, \lambda_v, \lambda_c$ control the relative weight of invariance, variance, and decorrelation terms, respectively.

Final Objective : The total training loss for X-JEPA is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{VICReg}} \quad (14)$$

This combined loss ensures latent semantic consistency, robustness to modality shifts, and stable feature decorrelation across sensor types. Detailed motivation with each component ablation given in *Supplementary Section B.*

4. Experiments

4.1. Datasets

We evaluate X-JEPA on two large-scale RS-CMIR benchmarks: BEN-14K [35] and fMoW [15, 16] encompassing diverse sensor modalities such as SAR, multispectral, and optical imagery. BEN-14K provides paired Sentinel-1 (SAR) and Sentinel-2 (multispectral) scenes labeled with 19 land cover categories, facilitating fine-grained cross-modal learning. fMoW contains high-resolution RGB images and corresponding Sentinel-2 multispectral data, each annotated with 62 semantic geospatial classes, supporting large-scale retrieval across sensors. Both datasets simulate heterogeneous sensor conditions and provide a valuable testbed for evaluating modality-invariant retrieval. Further dataset specifications, including preprocessing steps, sensor characteristics, and dataset statistics, are provided in *Supplementary Section A*.

4.2. Experimental Setup and Evaluation Protocol

X-JEPA is implemented using ViT-B/16 [17] as the shared backbone for both context and target encoders. These encoders operate in a Siamese configuration, independently processing disjoint masked views of the same input image. We use a patch size of 16×16 for all experiments, which provides the best trade-off between spatial resolution and semantic abstraction. A detailed analysis of patch size sensitivity is presented in *Supplementary Section B.2*. The prediction module adopts a lightweight transformer architecture with a 384-dimensional hidden size and 12 attention layers, consistent with the encoder's attention configuration to ensure representational symmetry. The cross-modal prediction is guided by the learnable shared queries and routed mask tokens passed through a combination of self-attention, cross-attention, and MLP layers, as detailed in Section 3.4. We train the model using the AdamW optimizer [29] with an initial learning rate of 10^{-4}, which is linearly warmed up to 10^{-3} over the first 15 epochs and then decayed to 10^{-6} via cosine annealing. Weight decay is progressively increased from 0.04 to 0.4. The target encoder weights are updated via an exponential moving average (EMA) [1], initialized at a momentum of 0.996 and annealed to 1.0 throughout training. A random disjoint masking strategy is applied separately to context and target views with a fixed masking ratio of 0.25. All models are trained with a batch size of 512 using mini-batch stochastic gradient descent. To prevent degenerate solutions and enhance representation diversity, we employ VICReg [5] as a feature regularization module with weights $[\lambda_v, \lambda_c, \lambda_i] = [25, 25, 1]$, enforcing variance, decorrelation, and invariance across modality-specific representations. For our proposed Prediction Space Alignment (PSA) loss, we empirically set $\lambda_{\text{PSA}} = 0.1$. To assess fair and comparative retrieval performance, following the standard CBIR protocols [22], we use the F1 score as the primary metric, computed over the top-5 retrieved neighbors for each query.

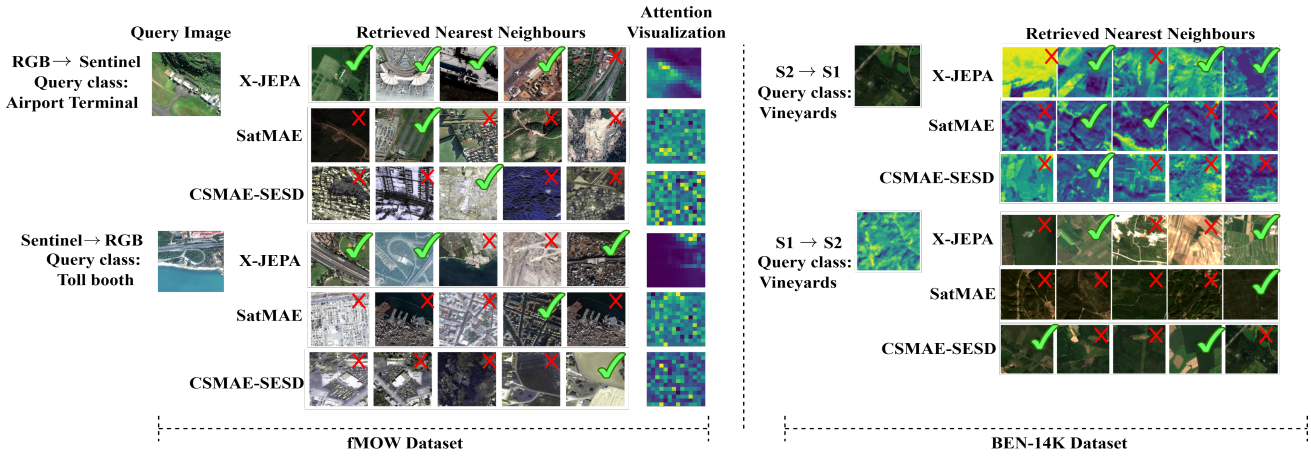


Figure 4. Qualitative cross-modal retrieval and attention visualizations from X-JEPA on the BEN-14K and fMoW datasets. The retrieved samples demonstrate the model's ability to recover semantically relevant scenes across heterogeneous sensor modalities.

4.3. Quantitative Retrieval Performance

We evaluate the effectiveness of X-JEPA on both unimodal and cross-modal retrieval tasks using the BEN-14K and fMoW benchmarks. Table 1 reports the F1-scores across all settings and compares our method against leading self-supervised models, including MAE [24], MAE-RVSA [40], SatMAE [16], SatMAE++ [31], Scale-MAE [32], CrossMAE [38], CSMAE-SESD [22], SkySense [21], CROMA [19], DeCUR [42] and REJEPA [14]. A consolidated view of performance across all retrieval tasks is illustrated in the radar chart of *Supplementary C.1*. All models are trained with ViT-B/16 backbones under the same self-supervised protocol for a fair comparison. To enable an equitable evaluation, we include CSMAE-SESD (Disjoint) as a baseline, as it shares the core design principle of disjoint masking between encoder branches, similar to our setup. For cross-modal evaluation, MAE-based baselines are adapted by independently pretraining on each modality and projecting features into a common embedding space.

On BEN-14K, X-JEPA achieves 72.98% F1 in S1→S1 and 82.37% in S2→S2, outperforming all previous approaches. More importantly, in cross-modal settings, it yields 61.23% and 63.65% in S1→S2 and S2→S1, respectively, an absolute improvement of 5–7% over SatMAE and nearly 15–23% over earlier MAE variants like MAE-RVSA and CSMAE-SESD. This demonstrates the strength of X-JEPA's predictive alignment and its ability to bridge semantic gaps between SAR and multispectral imagery. On fMoW, where intra-class variation and scene diversity pose challenges, X-JEPA also establishes a strong lead. It scores 75.26% in RGB→RGB and 77.22% in Sentinel→Sentinel, surpassing SkySense, CROMA, and CrossMAE. Notably, in the cross-modal RGB→Sentinel and Sentinel→RGB settings, it achieves 53.11% and 54.40%, respectively, outperforming the closest baseline by 2–4% and improving over CSMAE-SESD by more than 20%. These results affirm that our predictive feature-space learning mechanism generalizes effectively to diverse cross-sensors and scales.

Computational Cost Analysis: We also benchmark the efficiency of X-JEPA in terms of FLOPs and inference time on the BEN-14K dataset. Our model operates at approximately 10.8 GFLOPs per image with an average inference time of 20 ms, making it significantly faster than decoder-heavy counterparts like CrossMAE (19.6 GFLOPs, 77 ms) and even more efficient than MAE (12.1 GFLOPs, 43 ms), despite its added cross-modal semantic alignment capability. This demonstrates that X-JEPA achieves state-of-the-art retrieval accuracy without compromising computational efficiency.

4.4. Visualization Analysis

Figure 4 presents retrieval results and attention visualizations from X-JEPA on the BEN-14K and fMoW datasets. We benchmark our model against SatMAE [16] and CSMAE-SESD (Disjoint) [22]. In the top block, we present RGB→Sentinel retrievals with the query class *airport*. While SatMAE and CSMAE-SESD often retrieve patches with strong visual or structural similarity (e.g., open fields or road-like textures), they fail to capture the functional semantics of the target class. In contrast, X-JEPA retrieves scenes containing distinct airport infrastructure such as runways and terminals, showcasing its ability to align high-level semantic concepts across heterogeneous modalities. In the bottom block (Sentinel→RGB with the query class *toll booth*), both SatMAE and CSMAE-SESD again prioritize visually similar textures but miss semantically correct targets. X-JEPA, however, retrieves functionally relevant locations with clear road junctions and vehicular checkpoints, demonstrating its ability to reason beyond low-level appearance. The attention maps reveal that X-

Table 2. Ablation on Predictor depth on performance

Depth	S1→S1	S2→S2	S1→S2	S2→S1
6	51.89	76.78	54.04	51.85
8	67.11	78.76	57.44	55.14
10	70.31	81.44	54.09	53.81
12	**72.98**	**82.65**	**61.23**	**63.73**
14	72.05	81.92	60.77	62.38
16	71.20	81.21	60.19	61.53

Table 3. Ablation on Predictor Cross-attention module

Cross-Attn	Depth	S1→S1	S2→S2	S1→S2	S2→S1
✗	12	67.71	77.56	51.83	60.83
✓	4	57.43	79.45	48.36	53.19
	8	63.53	80.11	49.81	57.14
	10	71.57	81.62	54.05	60.34
	12	**72.98**	**82.65**	**61.23**	**63.73**

JEPA focuses on semantically meaningful regions (e.g., terminal buildings or booth structures) despite no explicit pixel-level supervision. Extended qualitative comparisons, category-wise attention maps and semantic explanations are provided in *Supplementary Sections C.1, C.2, and C.3 and C.4*

4.5. Ablation Study

We perform controlled ablation studies on BEN-14K to assess the contribution of core components in our framework, evaluated across both unimodal and cross-modal RS-CMIR settings.

Impact of Predictor Depth: Table 2 analyzes the effect of predictor depth on RS-CMIR. Increasing depth from 6 to 12 layers yields consistent F1 gains across both unimodal and cross-modal tasks (e.g., +21.1% for S1→S1, +11.8% for S2→S1). Beyond 12 layers (e.g., 14/16), we observe marginal improvement or slight degradation, indicating overfitting or reduced attention effectiveness.

Impact of Cross-Attention Module: Table 3 evaluates the role and depth of the cross-attention mechanism within the predictor module. Removing cross-attention (row 1) leads to a sharp drop in performance confirming its necessity for semantic alignment across modalities. With cross-attention enabled, increasing the number of attention layers from 4 to 12 consistently improves retrieval scores across all tasks. The 12-layer configuration achieves the best results. We empirically observed that depths beyond 12 (e.g., 14 and 16) offered negligible gains (<0.2%) while increasing computational cost; hence, they are omitted for conciseness.

Impact of Shared Queries Routing: Table 4 evaluates the role of shared semantic queries, which serve as modality-agnostic anchors in the predictor. Removing them results in consistent F1-score drops across all tasks (e.g., -7.1% for S1→S1 and S1→S2), indicating their importance in bridging modality gaps and guiding semantically consistent reconstruction across views.

Table 4. Ablation on Shared Queries strategy

Shared Queries	S1→S1	S2→S2	S1→S2	S2→S1
✗	70.86	78.95	54.05	54.73
✓	**72.98**	**82.65**	**61.23**	**63.73**

Table 5. Ablation on Masking Ratio of X-JEPA.

Masking Ratio	S1→S1	S2→S2	S1→S2	S2→S1
0.25	71.75	77.16	59.33	53.57
0.50	**72.98**	**82.65**	**61.23**	**63.73**
0.75	70.44	76.74	54.16	62.27

Table 6. Ablation on Loss function configurations

Loss Components	S1→S1	S2→S2	S1→S2	S2→S1
L2 only	68.85	76.22	52.46	60.34
L2 + L_{VICReg}	69.47	79.56	56.84	57.53
L2 + L_{PSA}	70.53	77.03	50.25	53.14
L2 + L_{PSA} + L_{VICReg}	**72.98**	**82.65**	**61.23**	**63.73**

Effect of Varying Masking Ratios: Table 5 analyzes the impact of different masking ratios. A balanced 0.50 ratio yields the best performance across tasks (82.65% for S2→S2 and 63.73% for S2→S1), offering sufficient learning signals without overwhelming the predictor. Lower ratios (e.g., 0.25) provide inadequate semantic challenges, while higher ones (e.g., 0.75) obscure essential context, especially harmful in cross-modal alignment.

Impact of Loss Function Configuration: Table 6 shows that using only L2 loss or a combination of L2 and L_{VICReg} yields limited retrieval performance, especially in cross-modal cases. Incorporating PSA loss improves unimodal scores but degrades cross-modal results due to insufficient regularization. When combined with VICReg, our full loss formulation significantly improves both unimodal (+9.8%) and cross-modal (+11.0%) F1-scores. This confirms that PSA enhances semantic alignment by structuring the latent space, while VICReg enforces stable feature distributions across modalities. A detailed analysis of PSA is provided in the *Supplementary Section B.3*.

Additional ablations on masking pattern comparisons (random vs. multiblock), sensitivity to PSA loss weight (λ_{PSA}), normalization strategies, and encoder patch size selection are provided in Supplementary Section B.

5. Takeaways

X-JEPA achieves superior performance across both unimodal and cross-modal tasks while remaining computationally efficient. Extension of this framework includes multi-query fusion, temporal retrieval, and large-scale geospatial pretraining across diverse sensors.

References

[1] Mahmoud Assran, Mathilde Caron, Ishan Misra, Piotr Bojanowski, Florian Bordes, Pascal Vincent, Armand Joulin, Mike Rabbat, and Nicolas Ballas. Masked siamese networks for label-efficient learning. In *European Conference on Computer Vision*, pages 456–473. Springer, 2022. 2, 6

[2] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15619–15629, 2023. 2, 3

[3] Guillaume Astruc, Nicolas Gonthier, Clement Mallet, and Loic Landrieu. Anysat: An earth observation model for any resolutions, scales, and modalities. *arXiv preprint arXiv:2412.14123*, 2024. 3

[4] Kumar Ayush, Burak Uzkent, Chenlin Meng, Kumar Tanmay, Marshall Burke, David Lobell, and Stefano Ermon. Geography-aware self-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10181–10190, 2021. 2

[5] Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. *arXiv preprint arXiv:2105.04906*, 2021. 2, 5, 6

[6] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *arXiv preprint arXiv:2404.08471*, 2024. 3

[7] Jules Bourcier, Gohar Dashyan, Karteek Alahari, and Jocelyn Chanussot. Learning representations of satellite images from metadata supervision. In *European Conference on Computer Vision*, pages 54–71. Springer, 2025. 2

[8] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9650–9660, 2021. 2

[9] Ushasi Chaudhuri, Biplab Banerjee, Avik Bhattacharya, and Mihai Datcu. Cmir-net: A deep learning based model for cross-modal retrieval in remote sensing. *Pattern recognition letters*, 131:456–462, 2020. 2

[10] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PMLR, 2020. 2

[11] Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 15750–15758, 2021. 2

[12] Yaxiong Chen, Jinghao Huang, Shengwu Xiong, and Xiaoqiang Lu. Fine aligned discriminative hashing for remote sensing image-audio retrieval. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–12, 2023. 2

[13] Shabnam Choudhury, Devansh Saini, and Biplab Banerjee. Boosting cross-modal retrieval in remote sensing via a novel unified attention network. *Neural Networks*, 180:106718, 2024. 2

[14] Shabnam Choudhury, Yash Salunkhe, Sarthak Mehrotra, and Biplab Banerjee. Rejepa: A novel joint-embedding predictive architecture for efficient remote sensing image retrieval. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 2373–2382, 2025. 2, 3, 6, 7

[15] Gordon Christie, Neil Fendley, James Wilson, and Ryan Mukherjee. Functional map of the world. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6172–6180, 2018. 6

[16] Yezhen Cong, Samar Khanna, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David Lobell, and Stefano Ermon. Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery. *Advances in Neural Information Processing Systems*, 35:197–211, 2022. 2, 6, 7

[17] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 6

[18] Zhengcong Fei, Mingyuan Fan, and Junshi Huang. A-jepa: Joint-embedding predictive architecture can listen. *arXiv preprint arXiv:2311.15830*, 2023. 3

[19] Anthony Fuller, Koreen Millard, and James Green. Croma: Remote sensing representations with contrastive radar-optical masked autoencoders. *Advances in Neural Information Processing Systems*, 36, 2024. 3, 6, 7

[20] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent-a new approach to self-supervised learning. *Advances in neural information processing systems*, 33:21271–21284, 2020. 2

[21] Xin Guo, Jiangwei Lao, Bo Dang, Yingying Zhang, Lei Yu, Lixiang Ru, Liheng Zhong, Ziyuan Huang, Kang Wu, Dingxiang Hu, et al. Skysense: A multi-modal remote sensing foundation model towards universal interpretation for earth observation imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27672–27683, 2024. 3, 6, 7

[22] Jakob Hackstein, Gencer Sumbul, Kai Norman Clasen, and Begüm Demir. Exploring masked autoencoders for sensor-agnostic image retrieval in remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 2, 3, 6, 7

[23] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9729–9738, 2020. 2

[24] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 16000–16009, 2022. 2, 6, 7

[25] Jinghao Huang, Yaxiong Chen, Shengwu Xiong, and Xiaoqiang Lu. Cross-modal remote sensing image-audio retrieval with adaptive learning for aligning correlation. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 2

[26] Yann LeCun. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62(1):1–62, 2022. 2, 3

[27] Haifeng Li, Yi Li, Guo Zhang, Ruoyun Liu, Haozhe Huang, Qing Zhu, and Chao Tao. Global and local contrastive self-supervised learning for semantic segmentation of hr remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–14, 2022. 2

[28] Weijie Li, Wei Yang, Tianpeng Liu, Yuenan Hou, Yuxuan Li, Zhen Liu, Yongxiang Liu, and Li Liu. Predicting gradient is better: Exploring self-supervised learning for sar atr with a joint-embedding predictive architecture. *ISPRS Journal of Photogrammetry and Remote Sensing*, 218:326–338, 2024. 3

[29] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. 6

[30] Oscar Manas, Alexandre Lacoste, Xavier Giró-i Nieto, David Vazquez, and Pau Rodriguez. Seasonal contrast: Unsupervised pre-training from uncurated remote sensing data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9414–9423, 2021. 2

[31] Mubashir Noman, Muzammal Naseer, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, and Fahad Shahbaz Khan. Rethinking transformers pre-training for multi-spectral satellite imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27811–27819, 2024. 2, 6, 7

[32] Colorado J Reed, Ritwik Gupta, Shufan Li, Sarah Brockman, Christopher Funk, Brian Clipp, Kurt Keutzer, Salvatore Candido, Matt Uyttendaele, and Trevor Darrell. Scale-mae: A scale-aware masked autoencoder for multiscale geospatial representation learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4088–4099, 2023. 2, 6, 7

[33] Ayumu Saito, Prachi Kudeshia, and Jiju Poovvancheri. Point-jepa: A joint embedding predictive architecture for self-supervised learning on point cloud. *arXiv preprint arXiv:2404.16432*, 2024. 3

[34] Geri Skenderi, Hang Li, Jiliang Tang, and Marco Cristani. Graph-level representation learning with joint-embedding predictive architectures. *arXiv preprint arXiv:2309.16014*, 2023. 3

[35] Gencer Sumbul, Arne De Wall, Tristan Kreuziger, Filipe Marcelino, Hugo Costa, Pedro Benevides, Mario Caetano, Begüm Demir, and Volker Markl. Bigearthnet-mm: A large-scale, multimodal, multilabel benchmark archive for remote sensing image classification and retrieval [software and data sets]. *IEEE Geoscience and Remote Sensing Magazine*, 9(3): 174–180, 2021. 6

[36] Gencer Sumbul, Markus Müller, and Begüm Demir. A novel self-supervised cross-modal image retrieval method in remote sensing. In *2022 IEEE International Conference on Image Processing (ICIP)*, pages 2426–2430. IEEE, 2022. 2, 3

[37] Xian Sun, Peijin Wang, Wanxuan Lu, Zicong Zhu, Xiaonan Lu, Qibin He, Junxi Li, Xuee Rong, Zhujun Yang, Hao Chang, et al. Ringmo: A remote sensing foundation model with masked image modeling. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–22, 2022. 2

[38] Maofeng Tang, Andrei Cozma, Konstantinos Georgiou, and Hairong Qi. Cross-scale mae: A tale of multiscale exploitation in remote sensing. *Advances in Neural Information Processing Systems*, 36, 2024. 6, 7

[39] Wei-Hsin Tseng, Hoàng-Ân Lê, Alexandre Boulch, Sébastien Lefèvre, and Dirk Tiede. Croco: Cross-modal contrastive learning for localization of earth observation data. *arXiv preprint arXiv:2204.07052*, 2022. 2

[40] Di Wang, Qiming Zhang, Yufei Xu, Jing Zhang, Bo Du, Dacheng Tao, and Liangpei Zhang. Advancing plain vision transformer toward remote sensing foundation model. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–15, 2022. 6, 7

[41] Yi Wang, Conrad M Albrecht, Nassim Ait Ali Braham, Lichao Mou, and Xiao Xiang Zhu. Self-supervised learning in remote sensing: A review. *IEEE Geoscience and Remote Sensing Magazine*, 10(4):213–247, 2022. 2

[42] Yi Wang, Conrad M Albrecht, Nassim Ait Ait Ali Braham, Chenying Liu, Zhitong Xiong, and Xiao Xiang Zhu. Decur: decoupling common & unique representations for multimodal self-supervision. 2023. 2, 6, 7

[43] Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A simple framework for masked image modeling. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9653–9663, 2022. 2

[44] Zhiqiang Yuan, Wenkai Zhang, Kun Fu, Xuan Li, Chubo Deng, Hongqi Wang, and Xian Sun. Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval. *arXiv preprint arXiv:2204.09868*, 2022. 2

[45] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *International conference on machine learning*, pages 12310–12320. PMLR, 2021. 2

[46] Lilu Zhu, Yang Wang, Yanfeng Hu, Xiaolu Su, and Kun Fu. Cross-modal contrastive learning with spatio-temporal context for correlation-aware multi-scale remote sensing image retrieval. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 3